

1. Define True/False. If false, give correct answer. 10
- i. Whether you write your own constructor or not, JVM always provides a default constructor for your class.

ii. Default members of a class can be accessed from subclass of different package.

iii. Interfaces may have instance variable.

iv. All methods in an *interface* must be static.

v. Interfaces can have abstract constructor.

vi. Super class object can be referenced by subclass variable.

vii. In java, a class can implement more than one interface.

viii. Abstract class can have *abstract* constructor

ix. An interface in Java can inherit more than one interface.

x. final methods cannot be overridden more than once.

2. What is the output of the following codes? Give explanation to each: (3+2) * 4

i)

<pre>public class Emp { static int bankVault; } public class TestDemo { public static void main(String args[]){ Emp emp1 = new Emp(); Emp emp2 = new Emp(); emp1.bankVault = 10; emp2.bankVault = 20; System.out.println(""+emp1.bankVault); System.out.println(""+emp2.bankVault); System.out.println(""+Emp.bankVault); } } public class Emp { static int bankVault; }</pre>	Ans:
--	------

ii)

<pre>public class TestJava { void show(){ try{ String s = null; System.out.println("1"); try{ System.out.println(s.length()); } catch(NullPointerException e){ System.out.println("inner"); } } } }</pre>	a) 1 inner 2 outer b) 1 inner 2
---	--

<pre> throw e; } System.out.println("2"); } catch (NullPointerException e){ System.out.println("outer"); } } } public static void main(String args[]){ TestJava ob = new TestJava(); ob.show(); } }</pre>	c) 1 inner outer d) Runtime Exception. e) None of the above
--	--

iii)

<pre>class Animal{ void jump(){ System.out.println("Animal"); } } class Dog extends Animal{ void jump(){ System.out.println("Dog"); } } class Horse extends Animal{ void jump(int a){ System.out.println("Horse"); } } public class Circus { public static void main(String args[]){ Animal dog = new Dog(); Horse horse = new Horse(); dog.jump(); horse.jump(); } }</pre>	a) Dog Horse b) Dog Animal c) Animal Animal d) Compilation Error e) None of the above
---	---

iv)

<pre>1. package A; 2. public class Alpha { 3. int a=1; 4. protected int b = 2; 5. public int c =3; 6. } 7. package A.B; 8. import A.Alpha; 9. public class Beta { 10. public static void main(String[] args){ 11. Alpha a = new Alpha(); 12. System.out.println("" + a.a); 13. System.out.println("" + a.b); 14. System.out.println("" + a.c); 15. } 16. }</pre>	a) 1 2 3 b) 1 2 3 c) Compilation fails with an error on line 11 d) Compilation fails with an error on line 12 e) Compilation fails with an error on line 13 f) Compilation fails with an error on line 14
---	---